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Optimal policy for an inventory system with power demand, backlogged shortages and production rate proportional to demand rate

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ABSTRACT

This article analyzes an inventory system for items with time-varying demand. The way by which demand occurs during the inventory cycle follows a power demand pattern. The production of items allows to add stock in the inventory during a replenishment period. The production rate is proportional to demand rate. Shortages are allowed and completely backlogged. Holding cost, shortage cost and order cost are the costs considered in the inventory model. The aim consists of the minimization of the total cost per inventory cycle. An efficient approach is developed to obtain the optimal scheduling period and the optimal reorder point. In addition, the economic lot size and the minimum cost of the inventory management is determined. Some numerical examples are presented for illustrating the proposed inventory model.

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1. Introduction

Inventory models give a variety of insights to find the optimal way to manage inventories of items. The classical economic order quantity (EOQ) model is one of the most popular and useful in inventory management. That model describes the trade-off between the holding cost and the order cost, deriving an economic order quantity that minimizes the inventory total cost per scheduling period. Since Harris (1913) presented the famous economic order quantity formula, many researchers have analyzed different inventory systems relaxing some of the assumptions of the basic model. The economic order quantity model assumes that the demand of items is constant. However, this assumption is not very common in practice, and could be more realistic to consider that demand depends on time.

The study of inventory models where it is allowed that demand rate varies with time has extended the field of inventory control and it has engaged the attention of researchers in the last decades. Thus, Resh et al. (1976) analyzed deterministic lot size systems with time-proportional demand. Donalson (1977) considered the situation of linearly time-varying demand and established an algorithm to determine the optimal policy. Barbosa and Friedman (1978) presented a continuous time inventory model with time-varying demand. Silver (1979) developed an approximate solution approach to reduce the computational effort needed in the procedure proposed by Donaldson. Dave and Patel (1981) studied a inventory model for deteriorating items with time proportional demand. Ritchie (1984) studied an inventory model with linear increasing demand. Mitra et al. (1984) presented a near-optimal procedure for adjusting the economic order quantity model for the cases of increasing or decreasing linear trend demand patterns. Goswani and Chaudhuri (1991) discussed an EOQ model with shortages, assuming a linear trend in demand and finite replenishment rate. Bose et al. (1995) developed an EOQ model with a linear positive trend, allowing backlogged shortages and deteriorating items. Chakrabarti and Chaudhuri (1997) developed an economic order quantity model that focused on time-varying demand. Zhao et al. (2001) analyzed the replenishment problem with decreasing demand, developing a new approach for that problem. Wen-Yang et al. (2002) presented a model for obtaining the exact optimal solution of inventory replenishment policy problem assuming a linear trend in demand. Wu (2002) discussed also an EOQ model with time-varying demand, considering deterioration and shortages. Goyal and Giri (2003) proposed an approach for determining replenishment intervals of an inventory item with linear decreasing demand rate. Yang et al. (2004) developed a procedure for replenishment of products with non-linear decreasing demand. Sakaguchi (2009) presented a model for a multiperiod inventory system with time-varying demand rate. Omar and Yeo (2009) studied a production system

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that satisfied a continuous time-varying demand for a finished product over a known and finite planning horizon. Mishra and Singh (2010) presented an inventory model for items with constant rate of deterioration and time dependent demand. Hsueh (2011) studied inventory policies for a manufacturing system with four phases of the product life cycle: introduction, growth, maturity, and decline. The mean demand rate of products varies according to the time in the product life cycle and is linear time dependent in the growth and decline periods. Maihami and Kamalabadi (2012) developed an inventory model for non-instantaneous deteriorating items, adopting a price and time dependent demand function. Pando et al. (2012) studied an profit maximization inventory model where the unit holding cost is a function of storage-time and the demand rate depends on stock-level. Recently, Pando et al. (2013) analyzed an economic lot-size inventory model with non-linear holding cost and dependent on time and quantity, assuming that demand rate depends on the stock level and, as consequence, varies with time.

The assumptions made about demand are decisive in the inventory systems. These usually determine the complexity of inventory management models and their utility in making business decisions. The different ways by which demand occurs during a period is referred to as demand patterns. Researchers have discussed different demand patterns fitting the stage of product cycle. Usually, the demand pattern is uniform and the demand rate is constant during all the period. However, there are other ways by which quantities are taken out of inventory to fill the customers' demand. One of these ways to remove items from the inventory is the called power demand pattern, which was discussed in Naddor (1966). That pattern allows to model the case of uniform demand and represents also other situations, where either a larger portion of demand occurs toward the beginning of the period or demand is mainly concentrated at the end of the period.

The power demand pattern is mainly observed for those products whose demand is not distributed uniformly through on the inventory cycle. We can identify real-life products demands that reflect situations where demand is higher at the beginning of the scheduling period. Thus, it occurs for cooked goods such as breads, cakes, sweets, pastries, prepared food, etc., because consumers want food products that was just finished. This type of demand can also occur in fish, fresh meat, fruits, vegetables, yogurts, etc., because sales are reduced when the expiry date approaches. However, there are other goods where the demand is higher at the end of the inventory cycle. This situation occurs for products like petrol, where demand increases when the good becomes scarce. Indispensable household items can fall into this category, such as oil, coffee, sugar, water, etc. Increases in their demand occur when the amount of inventory displayed starts to decrease because of its daily use.

From the point of view of business decisions, incorporating potential demand pattern may have interesting implications for management and inventory control within organizations. The analysis considering this demand pattern can improve the effectiveness and efficiency of different economic activities, determining the optimal policy and reducing management costs.

There are several papers where demand follows a power pattern. Thus, Naddor (1966) analyzed an order-level system with power demand pattern. Goel and Aggarwal (1981) developed an order-level inventory model with power demand for deteriorating items. Datta and Pal (1988) studied an inventory system with power demand and variable rate of deterioration. Lee and Wu (2002) presented an EOQ model assuming deterioration, shortages and power demand pattern. Dye (2004) extended the Lee an Wu' model to a general class with time-proportional backlogging rate. Singh et al. (2009) developed an economic order quantity model for perishable items with power demand pattern and partial backlogging. Rajeswari and Vanjikkodi (2011) studied an inventory model for deteriorating items where shortages are partially backlogged and demand follows a power pattern. Rajeswari and Vanjikkodi (2012) analyzed an inventory model for Weibull deteriorating items with power pattern time dependent demand. Mishra and Singh (2013) presented an EOQ model for perishable items with power demand in which shortages are allowed and partially backlogged.

In the above models where the power demand pattern is considered, the length of the inventory cycle is fixed. Thus, the average total cost per inventory cycle is a function that is depending on a variable. However, in the current work, the scheduling period is not fixed and the total cost per inventory cycle depends on two decision variables: the reorder point and the scheduling period. It implies that the solving of the inventory problem is a bit more difficult. Therefore, we will study replenishment policies in inventory systems with a power demand pattern and assuming a production rate proportional to demand rate. A mathematical formulation of the inventory system is presented where the different costs considered in the inventory model are specified. The objective of the inventory problem consists of determining the optimal scheduling period, the economic lot-size, the optimal reorder point and the minimum total inventory cost per unit time.

The rest of the paper is organized as follows. Section 2 introduces the assumptions and notation used throughout the paper. Section 3 presents the analysis of the inventory system. First, the average amount carried in inventory and the average shortage during the inventory cycle are calculated. Next, the costs associated with the inventory management are determined. Section 4 proposes a procedure to calculate the optimal solution for the inventory problem. Some special cases are analyzed in Section 5. Numerical examples are provided in Section 6 to illustrate the features of the proposed model. Finally, we present conclusions and some suggestions for future research.

2. Assumptions and notation

The inventory system will be assumed to have the following properties:

The planning horizon is infinite.

The behavior of the inventory system during the scheduling period T or length of each inventory cycle is assumed to be repeated continuously.

A single item is considered over the inventory cycle.

During the scheduling period T the demand size is denoted by d. Average demand is deterministic at a rate of $r=d/T$ quantity units per inventory cycle, but the way by which quantities are taken out of inventory depends on time when they are withdrawn. This way by which demand occurs during a period will be referred to as demand pattern.

The demand rate at time t follows a power demand rate and it is expressed as $D(t)=rt^{(1-n)/n}/nT^{(1-n)/n}$, with $0 \le t \le T$. Thus, the demand up to time t $(0 \le t \le T)$ is varying with time and is assumed to be $d(t/T)^{1/n}$, where n is the demand pattern index, with $0 < n < \infty$. Fig. 1 illustrates such power demand pattern for different values of the index n.

The inventory should be replenished whenever the inventory level is equal to or below s quantity units (s is the reorder point).

The replenishment quantity, production quantity or lot size Q is the quantity to be added to inventory.

The lead time is zero or negligible.

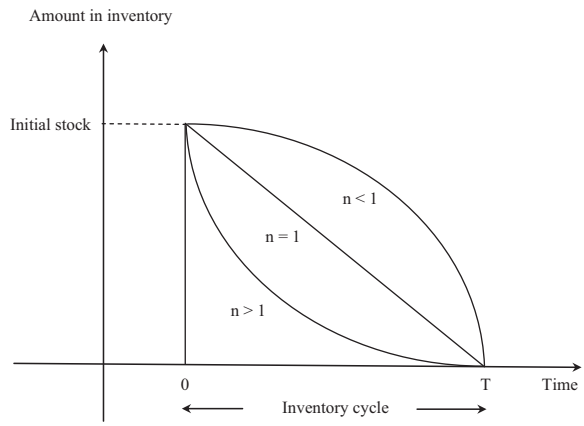

Fig. 1. Power demand pattern with index n. The nature of this demand pattern is entirely determined by n, the pattern index. If demand is uniform then the inventory declines linearly (index $n=1$). If a larger portion of demand occurs mainly toward the beginning of the period, then demand follows a pattern with index $n>1$. When a great part of demand occurs at the end of the inventory cycle, then the inventory is modeled by a power demand pattern with index $n<1$.

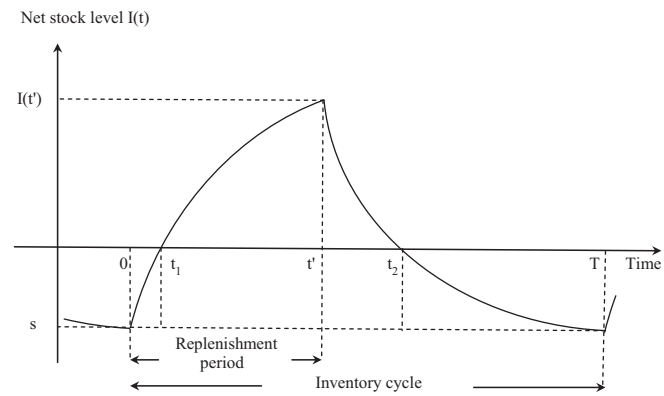

Fig. 2. Net stock level for a inventory system where demand follows a power pattern with index $n>1$ and the production rate is proportional to demand rate. Graphical representation of the fluctuations of the net inventory level when $-(\alpha-1)rT/\alpha \le s \le 0$. In the replenishment period $[0,t']$ the inventory increases due to the production rate is always greater than the demand rate. Next, the inventory decreases due to demand and drop to zero. Then shortage occurs until the end of the inventory cycle. Finally, when the maximum amount of backlogged demand is achieved, a new replenishment period begins and the inventory cycle is repeated continuously.

Replenishment is not instantaneous. There exist a replenishment period t' in which the product is replenished according with a production rate.
The production or replenishment rate is greater than the demand rate at any time during the replenishment period. The production rate $P(t)$ at any instant t $(0 \le t \le t')$ depends on the demand rate, and is given by $P(t)=\alpha D(t)$, with $\alpha>1$.
Shortages are allowed and backlogged.
The unit carrying cost or holding cost, the unit shortage cost or backlogging cost and the replenishment cost per order are constant.

We will use throughout the paper the following notation.

T=scheduling period or length of each inventory cycle.
s=reorder point.
Q=replenishment quantity or lot size.
t'=replenishment period.
d=demand size during the scheduling period.
r=average demand, that is, $r=d/T$.
h=carrying cost or holding cost (money/unit and unit time).
w=shortage cost or backlogging cost (money/unit and unit time).
A=replenishment cost (money/order).
$D(t)$=demand rate at time t $(0 \le t \le T)$.
$P(t)$=production rate at time t $(0 \le t \le t')$.
$I_1(s,T)$=average amount carried in inventory (units).
$I_2(s,T)$=average shortage in inventory (units).
$R(T)$=average number of replenishments per unit time.
$C_1(s,T)$=holding cost per unit time.
$C_2(s,T)$=shortage cost per unit time.
$C_3(T)$=replenishing cost per unit time.
$C(s,T)$=total cost per unit time of the inventory system.

The decision variables are the scheduling period T and the reorder point s. The input parameters of the inventory system are r, n, α, h, w and A.

The aim consists of determining the optimal scheduling period, the economic production quantity (EPQ) and the optimal reorder point that optimize the management of the inventory system described by the previous hypotheses. In the following section we present a mathematical model to study the inventory optimal policy.

3. Analysis of the inventory system

Let us consider a continuous review inventory system over an infinite-horizon with time-varying deterministic demand in which shortages are backlogged. Let $I(t)$ be the net stock (on hand—backorders) level at time t $(0 \le t \le T)$. At the beginning of the inventory cycle, the initial net stock is s units (see Fig. 2).

The replenishment period t' is the length of time during which the replenishment size is being added to inventory. Procurement through production starts just after $t=0$ and continue up to $t=t'$ when the stock attains a maximum level $I(t')$. Note that at any instant t, with $0 \le t \le t'$, the replenishment rate $P(t)$ must be larger than the demand rate $D(t)$. Thus, during the period $[0,t']$, inventories would have risen at a rate $P(t)-D(t)=(\alpha-1)rt^{(1-n)/n}/nT^{(1-n)/n}$.

Next, as the procurement is now stopped, the stock level declines up to $t=T$ due to demand and a new inventory cycle starts again.

Under above assumptions, the differential equations governing the system are

$$\frac{dI(t)}{dt}=(\alpha-1)\frac{rt^{(1-n)/n}}{nT^{(1-n)/n}}, \quad 0 \le t \le t'. \tag{1}$$

$$\frac{dI(t)}{dt}=-\frac{rt^{(1-n)/n}}{nT^{(1-n)/n}}, \quad t' \le t \le T. \tag{2}$$

The boundary conditions are $I(0)=I(T)=s$ (reorder point). The solutions of the above differential equations are

$$I(t)=s+(\alpha-1)rT\left(\frac{t}{T}\right)^{1/n}, \quad 0 \le t \le t'. \tag{3}$$

$$I(t)=s+rT-rT\left(\frac{t}{T}\right)^{1/n}, \quad t' \le t \le T. \tag{4}$$

At the point t' the net stock level given by (3) must be equal to the value $I(t')$ obtained by (4). Thus, equaling (3) to (4) at that

point, we have that the replenishment period t' is

$$t' = \frac{T}{\alpha^n} \tag{5}$$

Hence, Eq. (4) can be also written as

$$I(t) = s + \alpha rT\left(\frac{t'}{T}\right)^{1/n} - rT\left(\frac{t}{T}\right)^{1/n}, \quad t' \le t \le T. \tag{6}$$

The net stock level $I(t)$ is a T-periodic function and continuous defined on the unbounded interval $[0,\infty)$. In addition, $I(t)$ is an increasing function on $[0,t')$ and decreasing on $(t',T]$.

The total quantity demanded over $[0,T]$ is given by

$$\int_0^T D(t)dt = \int_0^T \frac{r}{n}\left(\frac{t}{T}\right)^{(1-n)/n} = rT \tag{7}$$

At the end of the inventory cycle a replenishment of size Q has been put into inventory. That lot-size is

$$Q = \int_0^{t'} P(t)dt = \alpha \int_0^{t'} \frac{r}{n}\left(\frac{t}{T}\right)^{(1-n)/n} = \alpha rT\left(\frac{t'}{T}\right)^{1/n} = rT \tag{8}$$

Therefore, the quantity scheduled for replenishment Q must fill the total demand rT throughout the inventory cycle.

The maximum amount in inventory is obtained when the replenishment is completely added to stock. Thus, the maximum amount in inventory is given by

$$I(t') = s + (\alpha-1)rT\left(\frac{t'}{T}\right)^{1/n} = s + \frac{(\alpha-1)rT}{\alpha} \tag{9}$$

3.1. The average amount carried in inventory and the average shortage during the inventory cycle

The inventory fluctuation of the system will depend on the relative values of the reorder point s and the maximum amount in inventory given in (9). There are three possible situations. If $s \ge 0$, there are no shortages and only inventories are carried in the system; if $-(\alpha-1)rT/\alpha \le s \le 0$, some inventories are carried and also shortages occur; and if $-(\alpha-1)rT/\alpha \ge s$, there are only shortages.

Both the average amount carried in inventory $I_1(s,T)$ and the average shortage $I_2(s,T)$ depend on the reorder point and scheduling period variables.

If $s \ge 0$, then only inventories are carried. Taking into account that the lot size is equal to total demand in the inventory cycle, that is $Q=rT$, the average amount carried in inventory is

$$I_1(s,T) = \frac{1}{T}\int_0^{t'}\left[s + (\alpha-1)\frac{rT}{\sqrt[n]{T}}t^{1/n}\right]dt + \frac{1}{T}\int_{t'}^{T}\left[s + Q - \frac{rT}{\sqrt[n]{T}}t^{1/n}\right]dt = \\ = \frac{1}{T}\left[(s+Q)T - \frac{Q^{n+1}T}{(n+1)(\alpha rT)^n} - \frac{rnT^2}{n+1}\right] = s + \frac{(\alpha^n-1)rT}{(n+1)\alpha^n} \tag{10}$$

In addition, in this case, no shortages occur and $I_2(s,T)=0$.

When $-(\alpha-1)rT/\alpha \ge s$, no inventories are carried and $I_1(S,T)=0$. The average shortage during the period T is given by

$$I_2(s,T) = \frac{-1}{T}\int_0^{t'}\left[s + (\alpha-1)\frac{rT}{\sqrt[n]{T}}t^{1/n}\right]dt - \frac{1}{T}\int_{t'}^{T}\left[s + Q - \frac{rT}{\sqrt[n]{T}}t^{1/n}\right]dt = \\ = \frac{-1}{T}\left[(s+Q)T - \frac{Q^{n+1}T}{(n+1)(\alpha rT)^n} - \frac{rnT^2}{n+1}\right] = \frac{(1-\alpha^n)rT}{(n+1)\alpha^n} - s \tag{11}$$

Finally, if $-(\alpha-1)rT/\alpha \le s \le 0$, some inventories are carried and shortages occur. Let t_1 be the time at which the inventory level reaches zero throughout the replenishment period, and let t_2 be the time at which the inventory level is zero during the period without replenishment. As $I(t_1)=0$, from (3), we have

$$t_1 = \frac{(-s)^n T}{[(\alpha-1)rT]^n} = \frac{(-s)^n}{r^n(\alpha-1)^n T^{n-1}} \tag{12}$$

Similarly, $I(t_2)=0$ and, from (5) and (6), we obtain

$$t_2 = \frac{(s+rT)^n T}{(rT)^n} = \frac{(s+rT)^n}{r^n T^{n-1}} \tag{13}$$

Fig. 2 illustrates the net inventory level of the system when $-(\alpha-1)rT/\alpha \le s \le 0$. In this case, the average amount carried in inventory is given by the expression

$$I_1(s,T) = \frac{1}{T}\int_{t_1}^{t'}\left[s + (\alpha-1)\frac{rT}{\sqrt[n]{T}}t^{1/n}\right]dt + \frac{1}{T}\int_{t'}^{t_2}\left[s + Q - \frac{rT}{\sqrt[n]{T}}t^{1/n}\right]dt$$

Taking into account (5), (12) and (13) and solving the above integral, we obtain

$$I_1(s,T) = \frac{(s+rT)^{n+1}}{(n+1)r^n T^n} - \frac{rT}{(n+1)\alpha^n} + \frac{(-s)^{n+1}}{(n+1)(\alpha-1)^n r^n T^n} \tag{14}$$

Now, we have to calculate the average shortage during the inventory cycle. This is given by

$$I_2(s,T) = \frac{-1}{T}\int_0^{t_1}\left[s + (\alpha-1)\frac{rT}{\sqrt[n]{T}}t^{1/n}\right]dt + \frac{-1}{T}\int_{t_2}^{T}\left[s + Q - \frac{rT}{\sqrt[n]{T}}t^{1/n}\right]dt$$

Considering (12) and (13) and solving the above integral, we have

$$I_2(s,T) = \frac{(-s)^{n+1}}{(n+1)(\alpha-1)^n r^n T^n} - s - \frac{rT}{n+1} + \frac{(s+rT)^{n+1}}{(n+1)r^n T^n} \tag{15}$$

Finally, the average number of replenishments per unit of time is always $R(T)=1/T$.

3.2. Costs in the inventory management

We introduce now the different costs considered in the inventory model. The holding cost is $C_1(s,T)=hI_1(s,T)$.

From (10), if $s \ge 0$, we have

$$C_1(s,T) = h\left[s + \frac{(\alpha^n-1)rT}{(n+1)\alpha^n}\right] \tag{16}$$

From (14), for $-(\alpha-1)rT/\alpha \le s \le 0$, the holding cost is

$$C_1(s,T) = h\left[\frac{(s+rT)^{n+1}}{(n+1)r^n T^n} + \frac{(-s)^{n+1}}{(n+1)(\alpha-1)^n r^n T^n} - \frac{rT}{(n+1)\alpha^n}\right] \tag{17}$$

Finally, if $-(\alpha-1)rT/\alpha \ge s$, no inventories are carried and we have $C_1(s,T)=0$.

Now, we calculate the shortage cost $C_2(s,T)=wI_2(s,T)$ for each situation. Firstly, if $s \ge 0$, no shortages occur and $C_2(s,T)=0$. From (15), when $-(\alpha-1)rT/\alpha \le s \le 0$, the shortage cost is

$$C_2(s,T) = w\left[\frac{(s+rT)^{n+1}}{(n+1)r^n T^n} + \frac{(-s)^{n+1}}{(n+1)(\alpha-1)^n r^n T^n} - s - \frac{rT}{n+1}\right] \tag{18}$$

If $-(\alpha-1)rT/\alpha \ge s$, from (11), the shortage cost is

$$C_2(s,T) = w\left[\frac{(1-\alpha^n)rT}{(n+1)\alpha^n} - s\right] \tag{19}$$

The replenishment cost is $C_3(T)=AR(T)=A/T$ for all the three cases.

The total cost per unit time is the sum of the three costs, that is, $C(s,T)=C_1(s,T)+C_2(s,T)+C_3(T)=hI_1(s,T)+wI_2(s,T)+AR(T)$. Thus, if $s \ge 0$, the total cost is given by

$$C(s,T) = h\left[s + \frac{(\alpha^n-1)rT}{(n+1)\alpha^n}\right] + A\frac{1}{T} \tag{20}$$

When $-(\alpha-1)rT/\alpha \le s \le 0$, the total cost per unit time of the system is given by

$$C(s,T)=h\left[\frac{(s+rT)^{n+1}}{(n+1)r^nT^n}+\frac{(-s)^{n+1}}{(n+1)(\alpha-1)^nr^nT^n}-\frac{rT}{(n+1)\alpha^n}\right]+w\left[\frac{(s+rT)^{n+1}}{(n+1)r^nT^n}+\frac{(-s)^{n+1}}{(n+1)(\alpha-1)^nr^nT^n}-s-\frac{rT}{n+1}\right]+A\frac{1}{T} \quad (21)$$

Finally, if $-(\alpha-1)rT/\alpha \ge s$, the total cost is given by

$$C(s,T)=w\left[\frac{(1-\alpha^n)rT}{(n+1)\alpha^n}-s\right]+A\frac{1}{T} \quad (22)$$

We now prove that the minimum of the function $C(s,T)$ must fall within the region characterized by the following condition: $-(\alpha-1)rT/\alpha \le s \le 0$. The minimum cannot be at $s \ge 0$, since the cost $C(s,T)$ is always greater than the cost $C(s=0,T)$. Similarly, the minimum of $C(s,T)$ cannot be in the region determined by the condition $-(\alpha-1)rT/\alpha \ge s$ since $C(s=-(\alpha-1)rT/\alpha,T)$ is always less than the cost $C(s,T)$. Thus, to obtain the solution of the inventory system, we have to find the minimum of the cost function $C(s,T)$ given by (20) within the range determined by $-(\alpha-1)rT/\alpha \le s \le 0$.

4. Optimal inventory policy

In this section we will analyze the optimal inventory policy. That policy can be found taking partial derivatives of the cost function given in (21). The partial derivative of $C(s,T)$ with respect to s is

$$\frac{\partial C(s,T)}{\partial s}=(h+w)\left[\frac{(s+rT)^n}{r^nT^n}-\frac{(-s)^n}{(\alpha-1)^nr^nT^n}\right]-w \quad (23)$$

and the partial derivative of $C(s,T)$ with respect to T is

$$\frac{\partial C(s,T)}{\partial T}=(h+w)\left[\frac{(s+rT)^n(rT-ns)}{(n+1)r^nT^{n+1}}-\frac{n(-s)^{n+1}}{(n+1)(\alpha-1)^nr^nT^{n+1}}\right]-\frac{hr}{(n+1)\alpha^n}-\frac{wr}{n+1}-\frac{A}{T^2} \quad (24)$$

Setting both derivatives equal to zero, we can find the optimal solution (s^*,T^*) of the inventory system. Thus, we have to solve the equations

$$(h+w)\left[\frac{(s+rT)^n}{r^nT^n}-\frac{(-s)^n}{(\alpha-1)^nr^nT^n}\right]-w=0 \quad (25)$$

$$(h+w)\left[\frac{(s+rT)^n(rT-ns)}{(n+1)r^nT^{n+1}}-\frac{n(-s)^{n+1}}{(n+1)(\alpha-1)^nr^nT^{n+1}}\right]-\frac{hr}{(n+1)\alpha^n}-\frac{wr}{n+1}-\frac{A}{T^2}=0 \quad (26)$$

Now, we consider a new variable x defined by the ratio between the amount of backorders and the total demand during the inventory cycle, that is,

$$x=\frac{-s}{rT} \quad (27)$$

Note that the searching region for obtaining the optimal policy is $-(\alpha-1)rT/\alpha \le s \le 0$, which is equivalent to the condition $0 \le x \le (\alpha-1)/\alpha$. Thus, the new variable x must vary inside the interval $[0,(\alpha-1)/\alpha]$.

Eq. (25) is equivalent to

$$(1-x)^n-\frac{x^n}{(\alpha-1)^n}-\frac{w}{h+w}=0 \quad (28)$$

and, from Proposition 1 (see Appendix), that equation has a unique solution x^* on the interval $(0,(\alpha-1)/\alpha)$.

That solution x^* can be found by using any numerical method as, i.e., the Newton–Raphson method (see, i.e., Stoer and Bulirsch (2002)).

Now, Eq. (26) is equivalent to

$$\frac{(h+w)r(1-x)^n(1+nx)}{(n+1)}-\frac{n(h+w)rx^{n+1}}{(n+1)(\alpha-1)^n}-\frac{hr}{(n+1)\alpha^n}-\frac{wr}{n+1}-\frac{A}{T^2}=0 \quad (29)$$

From (28), we have

$$\frac{x^n}{(\alpha-1)^n}=(1-x)^n-\frac{w}{h+w} \quad (30)$$

Substituting the above expression in (29) we obtain the equation

$$\frac{(h+w)r(1-x)^n}{n+1}+\frac{nwr}{n+1}x-\frac{hr}{(n+1)\alpha^n}-\frac{wr}{n+1}-\frac{A}{T^2}=0 \quad (31)$$

Now, solving Eq. (28) and substituting the obtained solution x^* in Eq. (31), we can obtain the optimal scheduling period T^*. That period is given by the formula

$$T^*=\sqrt{\frac{A}{r\left[((h+w)/(n+1))(1-x^*)^n+(nw/(n+1))x^*-(h/(n+1)\alpha^n)-(w/(n+1))\right]}} \quad (32)$$

From (27), the optimal reorder point is $s^*=-x^*rT^*$. The minimum cost per unit time of the inventory system is obtained substituting the values x^* and T^* in Eq. (21).

In addition, as $Q=rT$, the economic production quantity Q^* must be

$$Q^*=\sqrt{\frac{Ar}{((h+w)/(n+1))(1-x^*)^n+(nw/(n+1))x^*-(h/(n+1)\alpha^n)-(w/(n+1))}} \quad (33)$$

Finally, the following theorem proves that the point (s^*,T^*) is the optimal inventory policy of the system.

Theorem 1. *The solution (s^*,T^*) obtained solving Eqs. (28), (32) and (27) is the optimal inventory policy of the system.*

Proof. The second partial derivatives of $C(s,T)$ are

$$\frac{\partial^2C(s,T)}{\partial s^2}=(h+w)\left[\frac{n(s+rT)^{n-1}}{r^nT^n}+\frac{n(-s)^{n-1}}{(\alpha-1)^nr^nT^n}\right] \quad (34)$$

$$\frac{\partial^2C(s,T)}{\partial T^2}=(h+w)\left[\frac{ns^2(s+rT)^{n-1}}{r^nT^{n+2}}+\frac{n(-s)^{n+1}}{(\alpha-1)^nr^nT^{n+2}}\right]+\frac{2A}{T^3} \quad (35)$$

$$\frac{\partial^2C(s,T)}{\partial s\partial T}=(h+w)\left[\frac{n(-s)(s+rT)^{n-1}}{r^nT^{n+1}}+\frac{n(-s)^n}{(\alpha-1)^nr^nT^{n+1}}\right] \quad (36)$$

Let $H(s,T)$ be the Hessian of the function $C(s,T)$. It is given by

$$H(s,T)=\left[\frac{\partial^2C(s,T)}{\partial s^2}\right]\left[\frac{\partial^2C(s,T)}{\partial T^2}\right]-\left[\frac{\partial^2C(s,T)}{\partial s\partial T}\right]^2=(h+w)\left[\frac{n(s+rT)^{n-1}}{r^nT^n}+\frac{n(-s)^{n-1}}{(\alpha-1)^nr^nT^n}\right]\left[\frac{2A}{T^3}\right]>0 \quad (37)$$

Taking into account that the condition $s+rT>0$ is satisfied whenever $-(\alpha-1)rT/\alpha \le s \le 0$, then the second partial derivatives (34) and (35) are always positive and the Hessian H is also positive. Thus, the function $C(s,T)$ is a strictly convex function. Moreover, from (21), it follows that if $T\to\infty$ then it is not possible to reach the global minimum of the function $C(s,T)$ because the holding cost or the backlogging cost tends to infinite. Similarly, the ordering cost tends to infinite when $T\to 0$.Therefore, we conclude that the point (s^*,T^*) is a minimum point of the function $C(s,T)$. □

4.1. Algorithm

Summarizing the above results, a simple algorithm to find the optimal inventory policy is described below:

Step 1. Enter the input parameters of the inventory system.
Step 2. Calculate the unique solution x^* of Eq. (28) on the interval $[0,(\alpha-1)/\alpha]$ by using any numerical method.
Step 3. Determine the optimal scheduling period T^* by using the formula (32). Calculate the optimal reorder point given by the formula $s^*=-x^*rT^*$.
Step 4. Obtain the minimum cost $C^*=C(s^*,T^*)$ of the inventory system by using (21). Determine the economic production quantity Q^* from (33).

In the next section we will analyze the behavior of the above approach for some specific cases.

5. Special cases

In this section two special situations are analyzed. First, we will see how is the behavior of the approach proposed to obtain the optimal solution when the demand is uniform and, next, we will analyze the solution procedure when the replenishment is instantaneous.

5.1. Uniform demand

In the case that the index of the demand pattern is $n=1$, the demand rate and the production rate are constant, that is, $D(t)=r$ and $P(t)=\alpha r$. From (28), we can calculate the x-value. Now, we have to solve a single linear equation, whose solution is

$$x=\frac{(\alpha-1)h}{\alpha(h+w)} \tag{38}$$

Substituting that x-value in (32), we have the scheduling period

$$T=\sqrt{\frac{2A\alpha(h+w)}{r(\alpha-1)wh}} \tag{39}$$

From (33), the lot size is

$$Q=\sqrt{\frac{2Ar\alpha(h+w)}{(\alpha-1)wh}} \tag{40}$$

If we denote $p=\alpha r$, then the above expression can be written as

$$Q=\sqrt{\frac{2Ar(h+w)}{(1-r/p)wh}} \tag{41}$$

The formula (41) is the economic production quantity (EPQ) for the deterministic inventory system with uniform production rate and shortages completely backlogged (see Naddor (1966), Silver et al. (1998) and Zipkin (2000)). Thus, the model here analyzed extends to the inventory system previously cited.

5.2. Infinite production rate

We will analyze the behavior of the inventory system when the replenishment rate is not necessarily finite, as has been assumed in previous sections.

If we consider that α tends to ∞, then Eq. (28) reduces to

$$(1-x)^n=\frac{w}{h+w} \tag{42}$$

and the solution is

$$x=1-\sqrt[n]{\frac{w}{h+w}} \tag{43}$$

Substituting that x-value in (33), we have the optimal lot size

$$Q=\sqrt{\frac{Ar(n+1)}{nw(1-\sqrt[n]{(w/(h+w))})}} \tag{44}$$

That above formula is equal to the economic order quantity for an inventory system with a power demand pattern, instantaneous replenishment and shortages totally backlogged (see Sicilia et al. (2012)).

Thus, when $\alpha\to\infty$, the current system reduces to the previously cited system.

6. Numerical examples

Several numerical examples to illustrate the theoretical results are presented in the following paragraphs.

Example 1. Consider an inventory system with the same characteristics described in this work and assume the parameters are set as follows: $A=100$ \$ per order, $h=4$ \$ per unit and per unit time, $w=6$ \$ per unit and per unit time, $r=1200$ units per year, $\alpha=1.2$ and the index of the power demand pattern $n=2$.

From (28), we have to solve the equation

$$(1-x)^2-25x^2=\frac{6}{10} \tag{45}$$

The unique solution of the above equation inside the interval $0\le x\le(\alpha-1)/\alpha=1/6$ is $x^*=0.09399$. From (32), the scheduling period is $T^*=0.668965$ years$=244.172061$ days.

From (33), we obtain the economic replenishment size $Q^*=802.7575$ units.

From (27), the optimal reorder point is $s^*=-x^*rT^*=-75.451310$ units.

From (21), the minimum cost is $C^*=274.055404$ \$/year.

Finally, from (5), the replenishment period is $t'=T^*/\alpha^2=0.464559$ years$=169.563931$ days.

Example 2. Now, we consider the same input parameters, but only changing the index of the power demand pattern to $n=1/2$. Thus, Eq. (28) is

$$\sqrt{1-x}-\sqrt{5x}=\frac{3}{5} \tag{46}$$

Solving the above equation, we obtain the value $x^*=0.029654$. That value x^* is in the interval $(0, 1-1/\alpha)=(0,1/6)$.

From (32), the scheduling period is $T^*=Q^*/r=0.658703$ years$=240.426639$ days. From (33), the economic replenishment size is $Q^*=790.443743$ units.

The optimal reorder point is $s^*=-23.440159$ units and, from (21), the minimum cost is $C^*=303.626925$ \$/year. Considering Eq. (5), the replenishment period is $t'=T^*/\alpha^{1/2}=0.601311$ years$=219.478489$ days.

Some additional results are reported in Tables 1 and 2. We consider in Table 1 a power demand pattern index $n=3$ and the following input parameters: $r=1200$ units per year, $A=100$ \$ per order, and $h=4$ \$ per unit and per unit time. We choose for the production-rate parameter the following five values $\alpha=1.1$, 1.3, 1.5, 1.7 and 1.9. Four values are considered for the unit shortage cost w, that is, $w=0.1$, 0.5, 1.0 and 5.0.

Table 1 shows the results obtained for the optimal inventory policies considering several combinations of input data α and w.

Table 1
Optimal policies for inventory systems with power demand pattern and production rate proportional to demand rate, assuming several values for the production rate α and the unit shortage cost w.

Optimal inventory policy						
Input parameters: demand rate $r=1200$ units per year, ordering cost $A=100$ \$ per order, holding cost $h=4$ \$ per unit and time, and power demand index $n=3$.						
Production-rate parameter	Shortage cost per unit and time	x^*	s^*	T^*	Q^*	$C(s^*,T^*)$
$\alpha=1.1$	$w=0.1$	0.090007	−586.2918	5.4282	6513.8467	36.8486
	$w=0.5$	0.086659	−260.0917	2.5011	3001.3237	79.9647
	$w=1.0$	0.082970	−182.2317	1.8303	2196.3511	109.2724
	$w=5.0$	0.064156	−76.9536	0.9996	1199.4768	200.0881
$\alpha=1.3$	$w=0.1$	0.227568	−817.6352	2.9941	3592.9267	66.8006
	$w=0.5$	0.215648	−357.9292	1.3832	1659.7845	144.5961
	$w=1.0$	0.202449	−246.7011	1.0155	1218.5838	196.9506
	$w=5.0$	0.135125	−92.3058	0.5693	683.1144	351.3318
$\alpha=1.5$	$w=0.1$	0.327179	−912.2793	2.3236	2788.3185	86.0716
	$w=0.5$	0.304350	−394.1182	1.0791	1294.9505	185.3348
	$w=1.0$	0.279310	−267.2355	0.7973	956.7704	250.8435
	$w=5.0$	0.161603	−90.1421	0.4648	557.7996	430.2627
$\alpha=1.7$	$w=0.1$	0.402022	−966.1086	2.0026	2403.1238	99.8695
	$w=0.5$	0.366332	−411.5531	0.9362	1123.4428	213.6285
	$w=1.0$	0.328255	−274.5418	0.6970	836.3675	286.9553
	$w=5.0$	0.170823	−86.3391	0.4212	505.4304	474.8435
$\alpha=1.9$	$w=0.1$	0.459744	−1000.9186	1.8143	2177.1217	110.2366
	$w=0.5$	0.409842	−420.2422	0.8545	1025.3761	234.0603
	$w=1.0$	0.358947	−276.1700	0.6412	769.3892	311.9357
	$w=5.0$	0.174358	−83.4059	0.3986	478.3600	501.7140

Table 2
Optimal policies for inventory systems with power demand pattern and production rate proportional to demand rate, considering several values for the production rate parameter α and the power demand index n.

Optimal inventory policy						
Input parameters: demand rate $r=1200$ units per year, ordering cost $A=100$ \$, holding cost $h=4$ \$ per unit and time, and backlogging cost $w=5$ \$ per unit and time						
Index of the power demand pattern	Production-rate parameter	x^*	s^*	T^*	Q^*	$C(s^*,T^*)$
$n=0.2$	$\alpha=1.1$	0.001727	−2.4068	1.1614	1393.6332	172.2114
	$\alpha=1.3$	0.005142	−4.3594	0.7065	847.7928	283.0877
	$\alpha=1.5$	0.008506	−5.8474	0.5729	687.4440	349.1198
	$\alpha=1.7$	0.011818	−7.1519	0.5043	605.1672	396.5842
$n=0.5$	$\alpha=1.1$	0.018917	−20.8685	0.9193	1103.1600	217.5563
	$\alpha=1.3$	0.052391	−35.8783	0.5707	684.8184	350.4575
	$\alpha=1.5$	0.081199	−45.8768	0.4708	564.9924	424.7847
	$\alpha=1.7$	0.106348	−53.6988	0.4208	504.9348	475.3089
$n=0.8$	$\alpha=1.1$	0.033560	−36.2203	0.8994	1079.2584	222.3749
	$\alpha=1.3$	0.087651	−59.3511	0.5643	677.1322	354.4360
	$\alpha=1.5$	0.129529	−72.9582	0.4694	563.2589	426.0918
	$\alpha=1.7$	0.163048	−82.6140	0.4222	506.6850	473.6670
$n=1.0$	$\alpha=1.1$	0.040404	−44.0385	0.9083	1089.9540	220.1928
	$\alpha=1.3$	0.102564	−70.1645	0.5701	684.1044	350.8232
	$\alpha=1.5$	0.148148	−84.3272	0.4743	569.2092	421.6370
	$\alpha=1.7$	0.183007	−93.7249	0.4268	512.1384	468.6233
$n=1.5$	$\alpha=1.1$	0.051387	−58.2427	0.9445	1133.4132	211.7506
	$\alpha=1.3$	0.123697	−86.6098	0.5835	700.1772	342.7712
	$\alpha=1.5$	0.170756	−98.6110	0.4812	577.4964	415.5864
	$\alpha=1.7$	0.202918	−104.9295	0.4309	517.1028	464.1246
$n=2.0$	$\alpha=1.1$	0.057659	−67.4765	0.9752	1170.2688	205.0820
	$\alpha=1.3$	0.132912	−93.4119	0.5857	702.8100	341.4858
	$\alpha=1.5$	0.175842	−100.8813	0.4781	573.7044	418.3341
	$\alpha=1.7$	0.201163	−103.0885	0.4271	512.4624	468.3267
$n=3.0$	$\alpha=1.1$	0.064156	−76.9536	0.9996	1199.4768	200.0881
	$\alpha=1.3$	0.135125	−92.3058	0.5693	683.1144	351.3318
	$\alpha=1.5$	0.161603	−90.1421	0.4648	557.7996	430.2627
	$\alpha=1.7$	0.170823	−86.3391	0.4212	505.4304	474.8435
$n=5.0$	$\alpha=1.1$	0.068155	−78.4275	0.9589	1150.7232	208.5641
	$\alpha=1.3$	0.108902	−70.8316	0.5420	650.4156	368.9948
	$\alpha=1.5$	0.110740	−62.6922	0.4718	566.1204	423.9383
	$\alpha=1.7$	0.110879	−59.6096	0.4480	537.6096	446.4207

Fixed the unit shortage cost w, the value x^* (ratio between the amount of backorders and the total demand during the inventory cycle) increases as the production parameter α increases. However, the optimal scheduling period T^* and the optimal lot size Q^* decrease as the parameter α increases. Also, the minimum cost $C^*=C(s^*,T^*)$ increases as the production-rate parameter α increases.

In the same Table 1, fixed the production-rate parameter α, we can observe that if the unit shortage cost w increases then the value x^*, the optimal scheduling period T^* and the optimal lot size Q^* decrease. However, the optimal reorder point s^* and the minimum cost increase as the unit shortage cost w increases.

In Table 2, we show the optimal inventory policies when we vary the power demand pattern index $n=0.2$, 0.5, 0.8, 1.0, 1.5, 2.0, 3.0 and 5.0, and the production rate parameter $\alpha=1.1$, 1.3, 1.5 and 1.7. Fixed the index of the power demand pattern, the optimal scheduling period T^* and the economic lot size Q^* decrease as the production parameter α increases. However, the minimum cost increases as the production-rate parameter α increases. Also, if the production parameter α increases then x^* (ratio between the amount of backorders and the total demand during the inventory cycle) increases.

Fixed the production parameter α, from results shown in Table 2 we cannot find a standard pattern for the optimal inventory policy when the power demand index n increases. Thus, if $n \le 1$ then the value x^* and the total amount of backorders $-s^*$ increase as the power demand pattern index n increases. The optimal inventory cycle T^* and the economic lot size Q^* first decrease and then increase as the power demand index n increases. Also, the minimum cost C^* first increases and then decreases as the power demand index n increases.

However, fixed the production-rate parameter α and considering the results in Table 2, if $n>1$ then we cannot determine a behavior or tendency for the optimal inventory policy when n increases. Thus, when $n>1$, the value x^* and the total backorders $-s^*$ first increase and then decrease as the power demand index n increases. The optimal inventory cycle T^* and the economic lot size Q^* first increase, then decrease and finally increase again as the power demand index n increases. Also, the minimum cost C^* first decreases, then increases and finally decreases again as the power demand index n increases.

7. Conclusions and future research

In this work we develop an inventory system for a single item where the customers′ demand follows a power demand pattern. It is considered that shortages are allowed and backlogged, that is, shortages will be filled when the new replenishment arrives to the inventory system. The production rate to replenish the stock is considered to be proportional to demand rate. If demand increases then the production rate must also increase. Thus, a surplus is always added to stock and the inventory level increases during the replenishment period. When that period ends, the stock level in inventory declines up to the scheduling period is completed.

We have proposed a single approach to calculate the optimal inventory policy. The optimal solution obtained from the mathematical model is unique and it is easily determined by using an algorithm.

From the formulas of the optimal scheduling period, the economic production size, and the minimum inventory cost, it follows easily that if the order cost increases then the scheduling period, the economic production quantity and the inventory cost increase. Also, if demand rate increases then the economic production quantity increases, but the optimal scheduling period decreases. From the experimental results, it can be seen that if the production-rate parameter increases then the economic production quantity decreases, while the minimum inventory cost increases.

Future research could be to study inventory models with a power demand pattern and production rate proportional to demand rate, assuming that shortages are lost sales. Another line of research could be to analyze models of economic production quantity with a power demand pattern, in which it is assumed a deteriorate rate for items.

Appendix

Proposition 1. Eq. (28) has a unique real solution in the interval $(0,1-1/\alpha)$.

Proof. Let $y(x)$ be a real function defined on the interval [0,1] given by

$$y(x)=(1-x)^n-\frac{x^n}{(\alpha-1)^n}-\frac{w}{h+w} \tag{47}$$

That function $y(x)$ is a real-valued continuous function, differentiable and strictly decreasing on the interval (0,1) because its derivative is always negative, that is,

$$y'(x)=-n(1-x)^{n-1}-\frac{nx^{n-1}}{(\alpha-1)^n}<0 \tag{48}$$

In addition, we have that $y(0)=h/(h+w)>0$ and

$$y((\alpha-1)/\alpha)=-\frac{w}{h+w}<0 \tag{49}$$

Therefore, applying the Intermediate Value Theorem, there exist a point x^* in the interval $(0,1-1/\alpha)$ where $y(x^*)=0$. Finally, that point is unique because the function is decreasing on (0,1). □

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